

BiLD: Bi-directional Logits Difference Loss for Large Language Model Distillation

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Abstract

In recent years, large language models (LLMs) have shown exceptional capabilities across various natural language processing (NLP) tasks. However, such impressive performance often comes at the cost of an increased parameter size, posing significant challenges for widespread deployment. To address this issue, knowledge distillation (KD) provides a solution by transferring knowledge from a large teacher model to a smaller student model. In this paper, we explore the task-specific distillation of LLMs at the logit level. Our investigation reveals that the logits of fine-tuned LLMs exhibit a more pronounced long-tail distribution compared to those of vision models, with hidden "noise" in the long tail affecting distillation performance. Furthermore, existing logits distillation methods often struggle to effectively utilize the internal ranking information from the logits. To address these, we propose the **Bi-directional Logits Difference (BiLD)** loss. The BiLD loss filters out the long-tail noise by utilizing top- k teacher and student logits, and leverages the internal logits ranking information by constructing logits differences. To evaluate BiLD loss, we perform comprehensive experiments on 13 datasets with two types of LLMs. Our results show that the BiLD loss, with only the top-8 logits, outperforms supervised fine-tuning (SFT), vanilla KL loss, and five other distillation methods from both NLP and CV fields. Code is available at <https://github.com/fpcsong/BiLD>.

1 Introduction

The last few years have witnessed large language models (LLMs) risen to prominence, demonstrating remarkable proficiency in natural language understanding and generation. However, these capabilities come at the cost of an ever-increasing number of parameters. Due to constraints on computational resources, LLMs' formidable size limits

their democratization and widespread deployment. Knowledge distillation (KD), as a classic model compression method (Hinton et al., 2015), provides a solution for reducing model size while preserving performance. KD transfers knowledge from a large teacher model to a smaller student model, enhancing the latter's performance and enabling its lightweight deployment.

As an important branch of KD, logits distillation has gained popularity due to its straightforward application. The goal of logits distillation is to minimize the Kullback-Leibler (KL) divergence between the teacher and student logits. A significant portion of research on logits distillation has focused on vision models (Zhao et al., 2022; Yang et al., 2023a; Chi et al., 2023; Sun et al., 2024). However, the application of these methods to distill LLMs has yet to be thoroughly explored due to potential differences in structure, data distribution, and output space between vision and language models.

For LLMs, research on logits distillation is still emerging, with methods such as reverse KL (Tu et al., 2020; Lee et al., 2023; Gu et al., 2023) and those based on optimal transport metrics (Cui et al., 2024). However, in practical applications, the former suffers from the "mode-seeking" problem (Chan et al., 2022; Li and Farnia, 2023), while the latter is computationally too complex for open-source LLMs with billions of parameters.

In this paper, we investigate the characteristics of logits in LLMs. Compared to the limited output space of vision models, LLMs' output space comprises sequences of discrete tokens of potentially infinite length, making LLM logits significantly more complex. Furthermore, LLM logits exhibit a noticeable long-tail distribution, indicating a substantial portion of "noise" beyond a small amount of key knowledge. We also observe that in LLM text generation, common strategies like top- k sampling and top- p sampling are influenced

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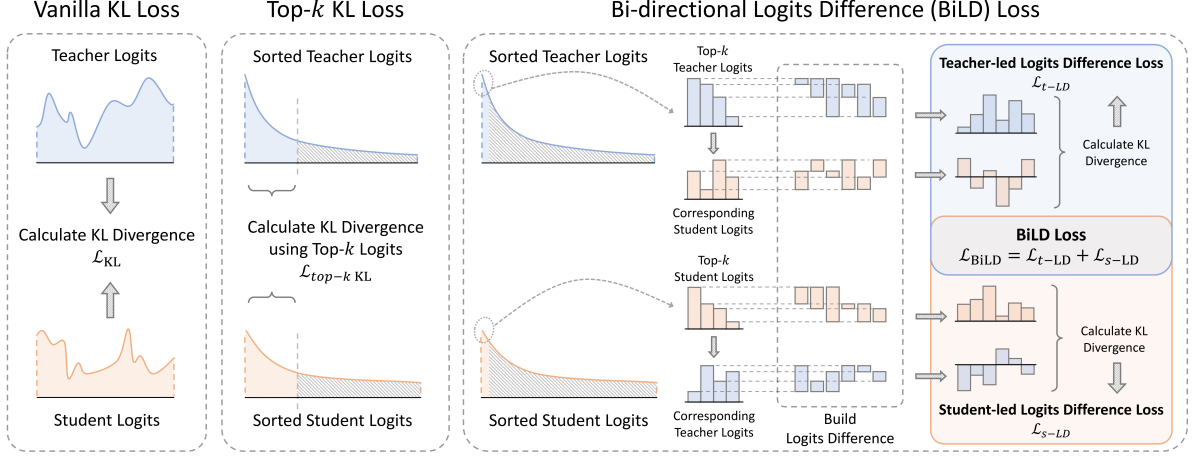


Figure 1: An illustration of vanilla KL, top- k KL and our BiLD loss. The vanilla KL loss directly calculates the KL divergence between teacher and student logits, whereas the top- k KL loss uses clipped logits instead of the full logits. In contrast to these methods, our BiLD loss computes KL divergence based on reconstructed logits differences. The logits difference is derived by calculating the pairwise differences between logit values. We construct two groups of logits differences and compute the KL divergence within each group as a loss: the top- k teacher logits and their corresponding student logits are used to calculate the teacher-led logits difference (t -LD) loss, while the top- k student logits and their corresponding teacher logits are used to calculate the student-led logits difference (s -LD) loss. The BiLD loss is the sum of these two losses.

by the internal ranking of logits when selecting output tokens. However, existing logits distillation methods often struggle to exploit this latent ranking information (Sun et al., 2024).

Motivated by the characteristics of logits and the underutilization of their internal ranking, we propose a novel loss function, called Bi-directional Logits Difference (BiLD) loss, for task-specific LLM distillation. BiLD loss emphasizes reducing long-tail noise and explicitly utilizes the ranking information in logits. It computes KL divergence based on reconstructed logits differences, which are obtained by calculating the internal pairwise differences of values from top- k teacher (student) logits and the corresponding student (teacher) logits. Our experiments show that BiLD loss, using only the top-8 logits, achieves state-of-the-art (SOTA) results across various NLP tasks. Our code is available at <https://github.com/fpcsong/BiLD>.

To conclude, we make the following contributions:

- We investigate the characteristics of LLMs’ logits, discussing their internal distribution and the significance of logits internal ranking.
- We propose the Bi-directional Logits Difference (BiLD) loss for logits distillation of LLMs. BiLD filters out "noise" in logits’ long-tail distribution while leveraging logits

ranking information to enhance performance. Our method can serve as an alternative to the vanilla KL loss in existing LLM distillation methods.

- To demonstrate the effectiveness of BiLD loss, we conduct comprehensive experiments on 13 NLP datasets using two open-source LLMs, BLOOM (BigScience Workshop, 2022) and Qwen1.5 (Bai et al., 2023). We evaluate various logits distillation methods from both CV and NLP domains. Experimental results show that our BiLD loss outperforms SFT, vanilla KL loss and five other methods using only the top-8 logits. Furthermore, our comparison of teacher and student logits shows that BiLD loss promotes better imitation of teacher’s primary behavior at the logit level.

2 Related Works

2.1 Logits Distillation

One representative approach of knowledge distillation is logits distillation, which transfers knowledge by minimizing the divergence of output logits (Jin et al., 2023). For vision models, there has been substantial research on logits distillation. Approaches like DKD (Zhao et al., 2022) and NKD (Yang et al., 2023b) decouple the target and non-target components of logits, applying weighting or regular-

ization. NormKD (Chi et al., 2023) dynamically customizes temperatures during the distillation process. However, the differences in structure, data, and output space between vision models and LLMs make it challenging to directly apply these methods to LLMs.

Recent research has introduced several logit distillation methods specifically for LLMs. Reverse KL (RKL) (Tu et al., 2020; Gu et al., 2023) has been used to mitigate the "mode-averaging" problem; however, it sometimes leads the student model towards "mode-seeking" behavior. DistiLLM (Ko et al., 2024) proposes mixing the logits distributions of the teacher and the student, but this introduces additional hyperparameters, increasing its complexity in practical applications. SinKD (Cui et al., 2024) replaces KL divergence with Sinkhorn Distance, but its computational demands pose challenges when applied to larger models.

Our work continues the paradigm of reducing the divergence of logits. However, unlike previous approaches, we calculate the divergence of logits differences instead of logits themselves. Our method focuses the model on the key knowledge in the teacher logits without introducing excessive hyperparameters.

2.2 Other Distillation Methods for LLMs

Previous works on distillation for LLMs extend beyond logits-based methods, primarily falling into two categories: white-box and black-box approaches (Yuan et al., 2024). White-box distillation (Gu et al., 2023; Liang et al., 2023; Agarwal et al., 2024) leverages the teacher’s internal representations and hidden states to facilitate knowledge transfer. However, these methods often rely on structural similarities between the teacher and student models. In contrast, black-box distillation only permits the student to access the teacher’s outputs. Current research in black-box distillation mainly focuses on learning from the teacher’s output texts (Sahu et al., 2023; Fu et al., 2023; Li et al., 2023). While BiLD can be classified as black-box distillation, it serves as an alternative to the vanilla KL divergence loss and can be easily integrated with white-box distillation methods.

3 Methods

3.1 Brief Review of Logits Distillation

Logits distillation calculates the divergence between the teacher’s and student’s logits as the op-

timization target. Consider a teacher model t and a student model s , both with a vocabulary size N . During the process of single token prediction, the teacher logits $\mathbf{z}^t$ and student logits $\mathbf{z}^s$ at a certain position can be represented as:

$$\begin{aligned}\mathbf{z}^t &= [z_1^t, z_2^t, \dots, z_N^t] \in \mathbb{R}^{1 \times N}, \\ \mathbf{z}^s &= [z_1^s, z_2^s, \dots, z_N^s] \in \mathbb{R}^{1 \times N}.\end{aligned}\quad (1)$$

Logits are the raw outputs of language models and cannot be directly used to calculate the loss. We process the logits into probabilities $\mathbf{p}^t$ and $\mathbf{p}^s$, where an element p_i from $\mathbf{p}^t$ or $\mathbf{p}^s$ represents the probability of the token at the i -th position being sampled as the output:

$$\begin{aligned}\mathbf{p}^t &= \frac{\exp(\mathbf{z}^t/T)}{\sum_{i=1}^N \exp(z_i^t/T)} \in \mathbb{R}^{1 \times N}, \\ \mathbf{p}^s &= \frac{\exp(\mathbf{z}^s/T)}{\sum_{i=1}^N \exp(z_i^s/T)} \in \mathbb{R}^{1 \times N},\end{aligned}\quad (2)$$

where T is the temperature during normalization. The vanilla KL divergence loss is defined as:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}[\mathbf{p}^t \parallel \mathbf{p}^s]. \quad (3)$$

By aligning the student’s logits with that of the teacher using vanilla KL loss, the student can imitate the teacher’s performance at the logit level, thereby facilitating knowledge transfer.

3.2 The Characteristics of LLMs’ Logits

Compared to vision models, LLMs have an output space consisting of infinitely long sequences of tokens, which makes their logits more complex. To compare the logit characteristics of vision models and LLMs, we conduct a toy experiment using ResNet-101 (He et al., 2016) and Qwen-4B (Bai et al., 2023). In this experiment, we randomly select five images and five sets of instructions from our test data as inputs for the vision and language models (details about images and instructions are provided in Appendix C). We use kurtosis to measure the extremity of logits’ long-tail distribution and calculate the proportion of top- k logit values. The experimental results are reported in Table 1. The kurtosis of text logits is 2-3 orders of magnitude higher than that of image logits, suggesting that text logits are much "sharper" than image logits. Given that text logits are much longer than image logits, the proportion of top- k logit values

Input Image / Text	Model	Kurtosis	Top- k logits percentage (%)			
			$k=8$	$k=64$	$k=512$	$k=1024$
cat.jpg	ResNet-101	975	99.540%	99.642%	99.993%	\
dogs.jpg		782	93.977%	98.433%	99.882%	\
lioness.jpg		995	99.904%	99.973%	99.999%	\
mushroom.jpg		914	99.756%	99.968%	99.998%	\
hat.jpg		906	83.982%	93.643%	99.646%	\
Instruction 1	Qwen-4B	135404	99.991%	99.996%	99.997%	99.998%
Instruction 2		46163	99.998%	99.998%	99.998%	99.998%
Instruction 3		79604	99.982%	99.990%	99.993%	99.994%
Instruction 4		50719	99.528%	99.604%	99.634%	99.651%
Instruction 5		116329	94.778%	94.826%	94.977%	95.081%

Table 1: The kurtosis and top- k proportion of image logits and text logits.

also indicates that text logit values are more peaked than those of image logits.

Moreover, previous logits distillation methods have not fully utilized the internal rank information of logits (Huang et al., 2022; Sun et al., 2024), even though this ranking information significantly affects LLMs’ generation performance. When LLMs generate text, two sampling strategies, top- k sampling and top- p sampling, are commonly used to control the diversity of the generated content. Top- k sampling controls the maximum length of the candidate tokens list, while top- p sampling filters tokens according to cumulative probability. The ranking of logit values impacts the selection process in both strategies, as higher-ranked tokens are more likely to be selected as candidates. Therefore, maintaining rank consistency will better assist the student in imitating the teacher’s generating patterns.

3.3 Bi-directional Logits Difference Loss

The Bi-directional Logits Difference (BiLD) loss is a novel optimization target for task-specific LLM distillation. It filters out the "noise" in the long-tail distribution of LLMs’ logits and constructs bi-directional differences that reflect the internal ranking of logits. Our goal is not for the student logits to fully match the teacher’s but for the student to effectively learn the key knowledge represented in the non-long-tail part. The detailed process of BiLD is shown in Figure 1.

3.3.1 Formal Definition

The BiLD loss consists of two components: the teacher-led logits difference (t -LD) loss and the student-led logits difference (s -LD) loss. Due to

the similarity between the two components, we explain the process by calculating the t -LD loss. First, we select the top- k teacher logits and sort them in descending order to build the teacher-led logits $\mathbf{z}_{\text{led}}^t$:

$$\mathbf{z}_{\text{led}}^t = [z_{i_1}^t, z_{i_2}^t, \dots, z_{i_k}^t] \in \mathbb{R}^{1 \times k}, \quad (4)$$

where the elements of $\mathbf{z}_{\text{led}}^t$ satisfy $z_{i_1}^t \geq z_{i_2}^t \geq \dots \geq z_{i_k}^t$. Then, we create the corresponding student logits $\mathbf{z}_{\text{cor}}^s$ by selecting the student logit values at the corresponding positions $[i_1, i_2, \dots, i_k]$:

$$\mathbf{z}_{\text{cor}}^s = [z_{i_1}^s, z_{i_2}^s, \dots, z_{i_k}^s] \in \mathbb{R}^{1 \times k}. \quad (5)$$

Next, we build the logits differences $\mathbf{d}_{\text{led}}^t$ and $\mathbf{d}_{\text{cor}}^s$ by calculating the internal pairwise value differences of $\mathbf{z}_{\text{led}}^t$ and $\mathbf{z}_{\text{cor}}^s$ respectively:

$$\begin{aligned} \mathbf{d}_{\text{led}}^t &= [z_{i_m}^t - z_{i_n}^t \mid 1 \leq m < n \leq k], \\ \mathbf{d}_{\text{cor}}^s &= [z_{i_m}^s - z_{i_n}^s \mid 1 \leq m < n \leq k], \end{aligned} \quad (6)$$

where both $\mathbf{d}_{\text{led}}^t$ and $\mathbf{d}_{\text{cor}}^s \in \mathbb{R}^{1 \times \frac{k(k-1)}{2}}$. Then we normalize $\mathbf{d}_{\text{led}}^t$ and $\mathbf{d}_{\text{cor}}^s$ into probabilities:

$$\begin{aligned} \mathbf{p}_{\text{led}}^t &= \frac{\exp(\mathbf{z}_{\text{led}}^t/T)}{\sum_{i=1}^{\frac{k(k-1)}{2}} \exp(z_{\text{led},i}^t/T)}, \\ \mathbf{p}_{\text{cor}}^s &= \frac{\exp(\mathbf{z}_{\text{cor}}^s/T)}{\sum_{i=1}^{\frac{k(k-1)}{2}} \exp(z_{\text{cor},i}^s/T)}. \end{aligned} \quad (7)$$

To obtain the teacher-led logits difference loss $\mathcal{L}_{t\text{-LD}}$, we calculate the KL divergence between $\mathbf{p}_{\text{led}}^t$ and $\mathbf{p}_{\text{cor}}^s$:

$$\mathcal{L}_{t\text{-LD}} = D_{\text{KL}} [\mathbf{p}_{\text{led}}^t \parallel \mathbf{p}_{\text{cor}}^s]. \quad (8)$$

The calculation of the s -LD loss is similar to that of the t -LD loss. The key difference is that the s -LD loss selects the top- k student logits $\mathbf{z}_{\text{led}}^s$ and extracts the corresponding teacher logits $\mathbf{z}_{\text{cor}}^t$. Based on these, we can sequentially calculate the logits differences $\mathbf{d}_{\text{led}}^s$ and $\mathbf{d}_{\text{cor}}^t$ as well as the probabilities $\mathbf{p}_{\text{led}}^s$ and $\mathbf{p}_{\text{cor}}^t$. The s -LD loss can be represented as:

$$\mathcal{L}_{s\text{-LD}} = D_{\text{KL}} [\mathbf{p}_{\text{cor}}^t \parallel \mathbf{p}_{\text{led}}^s]. \quad (9)$$

Finally, we obtain the BiLD loss:

$$\mathcal{L}_{\text{BiLD}} = \mathcal{L}_{t\text{-LD}} + \mathcal{L}_{s\text{-LD}}. \quad (10)$$

To aid comprehension, we outline the calculation process of the BiLD loss in Algorithm 1.

Algorithm 1 Calculation of BiLD Loss

Input: teacher logits $\mathbf{z}^t$, student logits $\mathbf{z}^s$, temperature T , hyperparameter k that controls the number of clipped logits

Output: the BiLD loss $\mathcal{L}_{\text{BiLD}}$

- 1: select top- k teacher logits $\mathbf{z}_{\text{led}}^t$ (Equation 4)
 - 2: select corresponding student logits $\mathbf{z}_{\text{cor}}^s$ (Equation 5)
 - 3: build the teacher and student logits differences $\mathbf{d}_{\text{led}}^t$ and $\mathbf{d}_{\text{cor}}^s$ (Equation 6)
 - 4: normalize differences to probabilities $\mathbf{p}_{\text{led}}^t$ and $\mathbf{p}_{\text{cor}}^s$ (Equation 7)
 - 5: calculate the teacher-led logits difference loss $\mathcal{L}_{t\text{-LD}}$ (Equation 8)
 - 6: calculate $\mathcal{L}_{s\text{-LD}}$ (Equation 9), generally following steps 1-5
 - 7: sum $\mathcal{L}_{t\text{-LD}}$ and $\mathcal{L}_{s\text{-LD}}$ to obtain $\mathcal{L}_{\text{BiLD}}$ (Equation 10)
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3.3.2 Explanation about the Utilization of Logits Ranking

The calculation of the logits difference (Equation 6) ensures that the student model learns the ranking information embedded in the teacher logits. We demonstrate this through an example based on the t -LD loss calculation. Since $\mathbf{z}_{\text{led}}^t$ satisfies $z_{i_1}^t \geq z_{i_2}^t \geq \dots \geq z_{i_k}^t$, it is guaranteed that every element in the teacher-led logits difference $\mathbf{d}_{\text{led}}^t$ is non-negative. For the corresponding student logits difference $\mathbf{d}_{\text{cor}}^s$, consider an element $d^s = z_{i_m}^s - z_{i_n}^s$. If $m > n$, then $d^s < 0$. In this case, the order

$z_{i_m}^s < z_{i_n}^s$ is inconsistent with the order in the teacher logits $z_{i_m}^t > z_{i_n}^t$. Therefore, the sign of the elements in the corresponding logits difference $\mathbf{d}_{\text{cor}}^s$ reflects whether the ranking of the teacher and student logits value pairs is consistent. When calculating $\mathcal{L}_{s\text{-LD}}$, this acts as a constraint, promoting the student logits to align their ranking order with the teacher logits.

4 Experiments

4.1 Datasets

We evaluate our BiLD loss on 13 NLP datasets: (1) 8 datasets from the SuperGLUE benchmark (Wang et al., 2019), including BoolQ (Clark et al., 2019), CB (De Marneffe et al., 2019), COPA (Roemmele et al., 2011), MultiRC (Khashabi et al., 2018), ReCoRD (Zhang et al., 2018), RTE (Giampiccolo et al., 2007), WiC (Pilehvar and Camacho-Collados, 2018) and WSC (Levesque et al., 2012); (2) 5 extra datasets used in previous works about model compression (Ma et al., 2023; Egiazarian et al., 2024), including: Arc-C, Arc-E (Clark et al., 2018), HellaSwag (Zellers et al., 2019), PIQA (Bisk et al., 2020) and WinoGrande (Sakaguchi et al., 2021). We observe that these datasets vary significantly in size (the visualization of dataset sizes is presented in Appendix B). Using small datasets alone for SFT and distillation would result in severe overfitting. To prevent unreliable experimental results, we use these datasets collectively for SFT and distillation, then evaluate each separately.

4.2 Baselines

We compare BiLD loss with seven baselines: (1) supervised fine-tuning (SFT), where all parameters are adjusted during adaptation to downstream tasks; (2) vanilla KL loss; (3) vanilla KL loss with only top- k logits (short as top- k KL), to demonstrate the performance improvements from noise filtering; (4) three logits distillation methods for vision models, including DKD (Zhao et al., 2022), NKD (Yang et al., 2023b), and NormKD (Chi et al., 2023); (5) Reverse KL loss (RKL) used in MiniLLM (Gu et al., 2023), which has been proven to enhance distillation performance on LLMs.

4.3 Implementation Details

We conduct experiments using the BLOOM and Qwen1.5 (abbreviated as Qwen) models, which are available in various sizes. Specifically, we em-

Model	Method	Arc-C (Acc.)	Arc-E (Acc.)	boolQ (Acc.)	CB (Acc.)	COPA (Acc.)	HellaSwag (Acc.)	MultiRC (F1a/EM)	PIQA (Acc.)	ReCoRD (F1/Acc.)	RTE (Acc.)	WiC (Acc.)	WinoGrande (Acc.)	WSC (Acc.)	Avg.
BLOOM-7B	Teacher	50.84	68.95	85.26	89.29	81.00	76.08	81.36/40.82	74.92	79.87/78.50	83.03	72.41	71.51	65.38	72.15
BLOOM-3B	SFT	44.15	61.75	84.04	87.50	67.00	57.00	77.09/36.20	70.84	76.05/74.59	78.34	69.75	69.69	64.42	66.56
	Vanilla KL	49.50	68.07	84.50	87.50	76.00	72.60	78.89/36.52	74.27	79.81/78.32	81.59	71.94	70.96	74.04	71.21
	RKL	50.50	68.42	84.62	87.50	80.00	72.20	78.95/36.41	74.48	79.63/78.13	82.31	72.57	71.35	68.27	71.29
	DKD	49.50	69.82	85.26	91.07	80.00	71.54	77.84/35.68	73.01	79.09/77.65	79.42	73.20	70.96	66.35	71.04
	NKD	50.17	67.19	84.01	92.86	79.00	72.68	79.69/37.67	73.50	78.50/77.09	81.23	71.32	72.06	66.35	71.16
	NormKD	48.16	67.54	85.35	89.29	79.00	70.57	77.19/35.57	71.82	78.44/76.98	80.87	72.88	70.48	68.27	70.52
	Top- <i>k</i> KL	47.49	68.25	84.19	87.50	77.00	72.75	79.39/37.67	74.59	79.40/78.01	82.67	72.10	70.80	64.42	70.57
	BiLD (ours)	49.83	67.54	84.86	91.07	80.00	72.10	79.49/37.78	73.61	79.96/78.57	82.67	72.88	71.98	71.15	71.85
BLOOM-1B	SFT	34.78	53.86	80.76	87.50	64.00	37.39	73.18/30.12	65.72	72.04/70.59	73.65	67.71	67.40	64.42	61.38
	Vanilla KL	45.48	64.39	83.67	87.50	73.00	65.31	77.66/33.37	70.95	77.11/75.67	77.62	68.03	68.43	68.27	67.82
	RKL	45.48	65.44	83.43	85.71	74.00	65.70	76.63/32.95	70.78	77.51/76.10	79.42	70.69	68.27	64.42	67.88
	DKD	42.47	64.56	84.10	85.71	72.00	63.72	75.49/31.79	69.48	75.78/74.46	79.78	71.79	68.98	69.23	67.55
	NKD	43.14	60.88	82.75	89.29	68.00	63.53	76.94/34.84	70.73	75.31/73.87	77.62	69.44	69.30	61.54	66.53
	NormKD	42.81	61.05	83.82	83.93	69.00	62.80	74.13/30.75	67.74	74.49/72.95	77.62	69.91	67.80	65.38	65.81
	Top- <i>k</i> KL	49.50	62.11	83.06	89.29	74.00	65.72	78.30/34.73	71.22	77.28/75.89	77.98	70.22	69.30	60.58	67.97
	BiLD (ours)	44.48	62.98	83.39	91.07	77.00	64.84	78.37/35.78	72.20	77.23/75.93	80.14	70.53	69.30	68.27	68.92
Qwen-4B	Teacher	68.23	81.40	87.43	96.43	89.00	86.30	85.85/51.63	82.10	82.59/81.10	87.73	72.73	80.82	74.04	79.92
Qwen-1.8B	SFT	52.17	73.86	83.88	91.07	86.00	72.58	79.95/39.66	75.90	77.37/76.05	84.12	71.79	72.06	61.54	72.36
	Vanilla KL	55.52	74.74	85.60	96.43	86.00	77.74	79.46/36.52	76.66	79.24/36.52	85.56	69.59	75.14	64.42	73.98
	RKL	50.84	76.14	85.14	94.64	87.00	77.85	79.52/39.14	76.39	79.49/77.98	84.48	71.47	76.64	69.23	74.38
	DKD	51.84	77.02	85.75	98.21	85.00	76.90	80.56/39.77	74.54	77.91/76.18	84.48	71.16	76.56	67.31	74.21
	NKD	51.84	73.33	84.53	92.86	88.00	77.49	81.98/42.18	76.61	79.03/77.58	84.12	70.85	74.98	66.35	73.90
	NormKD	52.84	76.49	85.26	96.43	85.00	77.24	80.81/40.50	74.76	78.22/76.48	85.92	70.53	76.87	70.19	74.50
	Top- <i>k</i> KL	53.85	76.14	85.93	96.43	82.00	77.99	81.81/41.03	76.71	80.08/78.71	83.39	71.32	75.85	67.31	74.36
	BiLD (ours)	54.85	73.16	84.53	96.43	88.00	77.56	81.49/42.92	77.97	79.87/78.56	85.56	72.10	76.01	68.27	75.07
Qwen-0.5B	SFT	37.46	62.11	80.40	87.50	77.00	46.71	74.24/28.54	68.44	71.19/69.79	77.26	66.30	69.38	59.62	63.88
	Vanilla KL	43.14	63.68	81.74	85.71	78.00	66.73	75.97/29.07	71.87	72.55/70.91	79.78	70.53	71.35	60.58	67.16
	RKL	46.49	64.39	81.53	87.50	79.00	67.06	75.37/29.38	71.16	71.46/69.55	82.31	69.91	70.80	58.65	67.52
	DKD	40.80	62.98	82.66	82.14	77.00	61.03	72.35/26.55	66.87	65.68/63.20	81.59	70.06	70.64	61.54	65.16
	NKD	41.14	63.86	82.42	94.64	78.00	68.30	79.33/36.20	73.01	74.81/73.35	82.31	67.40	72.22	71.15	69.54
	NormKD	41.14	61.40	82.72	83.93	77.00	62.31	74.13/29.07	68.55	67.17/64.79	82.31	71.16	71.43	62.50	66.02
	Top- <i>k</i> KL	43.14	65.79	82.39	94.64	77.00	68.58	78.83/35.89	71.82	74.30/72.95	82.31	69.28	73.24	62.50	69.19
	BiLD (ours)	41.81	67.54	83.43	96.43	78.00	68.99	79.72/37.78	73.34	75.22/73.94	81.59	69.75	72.22	74.04	70.68

Table 2: The overall performance of various distillation methods and SFT baselines, with best results shown in **bold**. When choosing the best results and calculating the Average Accuracy (Avg.), we use EM score for the MultiRC dataset and Accuracy for the ReCoRD dataset. The instruction templates for each dataset are listed in Appendix D.

ploy BLOOM-7B and Qwen-4B as teacher models. For student models, we select BLOOM-3B and BLOOM-1B from the BLOOM series, and 1.8B and 0.5B versions from Qwen.

We perform three epochs of SFT on each teacher model and eight epochs of distillation for each student. Both SFT and distillation are conducted with a batch size of 64 and a micro batch size of 2, using the full dataset. We employ a cosine scheduler with an initial learning rate of $1e-5$ for SFT and $2e-5$ for distillation. The warm-up steps are set to 64. During SFT, we utilize the cross entropy loss. For the different distillation methods we tested, all parameters, except for temperature, are set to their default values. Due to the computational complexity of some distillation methods, we use the vanilla

KL loss for the instruction part to expedite the distillation process, and apply different distillation losses to the output part. The temperature T for all loss functions is 3. For the top- k KL loss, we set $k=1024$, and for our proposed BiLD loss, we set $k=8$.

All our experiments are carried out on 8 NVIDIA A100 GPUs. To reduce memory usage, we employ DeepSpeed (Rasley et al., 2020) during both SFT and distillation processes, along with gradient checkpointing and BFLOAT16 mode (Kalamkar et al., 2019). We have not explored the minimum memory requirements. However, in practice, experiments involving all methods except DKD (Zhao et al., 2022), NKD (Yang et al., 2023b), and NormKD (Chi et al., 2023) can be conducted with

half of the computational resources. During the evaluation, we employ vLLM (Kwon et al., 2023) for faster inference. The evaluation can be performed with a single NVIDIA A100 GPU. More implementation details can be found in our repository.

5 Results and Analysis

5.1 Main Results

We report the experimental results on all 13 datasets in Table 2. Across all four sets of distillation, the BiLD loss achieves the highest average accuracy, outperforming SFT, vanilla KL, and the other five methods we tested. In the distillation from Qwen-4B to 0.5B, the BiLD loss showed a significant improvement in average accuracy, surpassing the vanilla KL loss by 3.52%. This improvement is also observed in the distillation from Qwen-4B to 1.8B and from BLOOM-7B to 1B, with improvements of 1.09% and 1.10% over the vanilla KL loss respectively. A notable case is the distillation from BLOOM-7B to 1B, where the student using vanilla KL loss can easily match the teacher’s performance. In this scenario, our BiLD loss still maintained a consistent advantage, with an average increase of 0.64% over the vanilla KL loss. In contrast, other methods show only marginal improvements or even declines in performance. The robust performance of the BiLD loss across various distillation scenarios underscores its superiority and effectiveness.

5.2 Analysis of the Effectiveness of Clipping Logits

The experimental results in Table 2 indicate that, in three distillation scenarios, simply clipping the full logits to the top- k logits improves the performance of the KL loss. This suggests that filtering out the noise in the logits’ long tail distribution can be a practical and straightforward approach to enhancing distillation performance. Our statistics show that the top-1024 logits cover over 99% of the probability in both Qwen-4B and BLOOM-7B teachers. For computational simplicity, we set $k=1024$ for the top- k KL loss to verify that excluding the long-tail distribution of logits can improve distillation results.

5.3 Analysis of Performance at the Logit Level

To demonstrate the performance of different distillation methods at the logit level, we introduce a new metric, top- k overlap (overlap@ k). Con-

Model	Method	Overlap@1	Overlap@8
BLOOM-3B	SFT	74.89	44.61
	Vanilla KL	82.51	54.64
	RKL	82.31	54.64
	DKD	74.00	52.39
	NKD	82.11	53.25
	NormKD	48.80	36.95
	Top- k KL	81.67	55.73
	BiLD	81.72	56.57
BLOOM-1B	SFT	74.40	40.71
	Vanilla KL	80.82	51.91
	RKL	80.71	51.58
	DKD	75.44	48.83
	NKD	79.59	50.01
	NormKD	73.56	42.70
	Top- k KL	80.20	50.87
	BiLD	81.21	52.86
Qwen-1.8B	SFT	93.30	53.28
	Vanilla KL	94.35	68.02
	RKL	94.31	67.93
	DKD	94.09	67.01
	NKD	94.02	65.01
	NormKD	94.26	68.32
	Top- k KL	94.43	67.55
	BiLD	94.39	70.97
Qwen-0.5B	SFT	91.67	47.29
	Vanilla KL	92.72	61.81
	RKL	92.54	61.65
	DKD	91.50	56.62
	NKD	92.88	59.11
	NormKD	91.76	58.16
	Top- k KL	93.11	64.00
	BiLD	93.23	68.58

Table 3: The top-1 and top-8 overlap of different distillation methods on 4 distillation settings.

sider an instruction I represented as a sequence of tokens. We denote the output tokens generated by the teacher with I as A^t , and the concatenated sequence of tokens as $C^t = I \oplus A^t$. The logits sequence for the teacher’s output part can be represented as $\mathbf{Z}^t = [\mathbf{z}_1^t, \mathbf{z}_2^t, \dots, \mathbf{z}_M^t]$, where M is the length of A^t . The element $\mathbf{z}_i^t$ within $\mathbf{Z}^t$ is the logits at the i -th position of the teacher’s output part. By feeding the whole C^t into the student, we denote the student logits sequence corresponding to the positions of A^t as $\mathbf{Z}^s = [\mathbf{z}_1^s, \mathbf{z}_2^s, \dots, \mathbf{z}_M^s]$. Consequently, we define the top- k overlap as:

$$\text{overlap}@k = \frac{1}{M} \sum_{i=1}^M \frac{\text{topk}(\mathbf{z}_i^t) \cap \text{topk}(\mathbf{z}_i^s)}{k}, \quad (11)$$

where $\text{topk}(\cdot)$ is a function to select tokens corresponding to the top- k logit values. The metric overlap@ k measures the average degree of over-

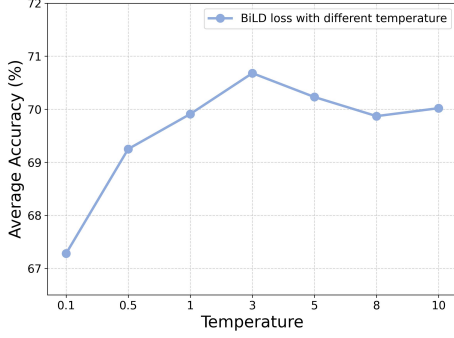


Figure 2: Ablation study of model temperature.

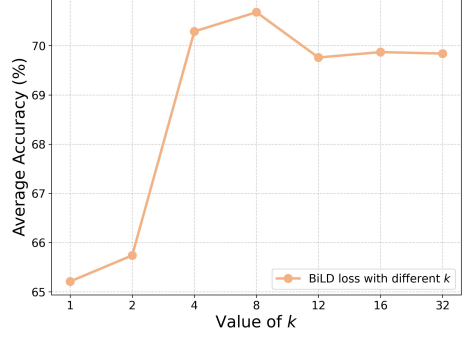


Figure 3: Ablation study of k values in BiLD loss.

lap for the top- k logits corresponding tokens at the same positions in $\mathbf{Z}^t$ and $\mathbf{Z}^s$. Specifically, overlap@1 evaluates whether the tokens with the highest logit values from the teacher and student outputs match at each position. This metric can measure the efficacy of LLMs in greedy search mode, where LLMs generate text based on the token with the highest probability. For $k > 1$, $\text{overlap}@k$ calculates the ratio of overlapping tokens corresponding to the top- k logits from both student and teacher at each position, reflecting how well the student imitates the important parts of teacher logits. From another perspective, overlap@1 measures the performance of models in scenarios where there is only one correct answer, while $\text{overlap}@k (k > 1)$ reflects the degree of similarity between the student and teacher responses in open-ended scenarios.

According to the results in Table 3, our proposed BiLD loss notably enhances overlap@8 while maintaining a competitive overlap@1 . Compared to other methods, students trained with BiLD loss better imitate the teacher’s primary behaviors at the logit level, indicating that BiLD loss helps student logits align with the important part of teacher logits.

5.4 Ablation Study

5.4.1 Impact of Temperature

To understand the impact of temperature during the distillation of BiLD loss, we vary the temperature parameter $T \in \{0.1, 0.5, 1, 3, 5, 8, 10\}$ while keeping other hyperparameters and model architectures constant. The experimental results, as depicted in Figure 2, indicate that lower temperatures significantly degrade the performance of BiLD loss. We choose $T=3$, which yields the best performance, for our distillation experiments.

5.4.2 Impact of the k Value in BiLD Loss

The hyperparameter k controls the length of clipped logits in BiLD loss. We experiment with $k \in \{1, 2, 4, 8, 12, 16, 32\}$ and evaluate the distillation results using average accuracy as well as top-1, top-8, and top-32 overlap. We report the results in Figure 3 and Table 4. Smaller k values ($k \in \{1, 2\}$) lead to overly short logits, resulting in poor performance. As k increases, both average accuracy and overlap@1 rise and then stabilize, while significant improvements can be seen in overlap@8 and overlap@32 . However, higher k values lead to increased computational costs. Considering the trade-off between computation time and performance, we select $k=8$ for BiLD loss in our experiments.

top- k	Overlap@1	Overlap@8	Overlap@32
$k=1$	91.93	49.57	38.91
$k=2$	91.97	49.60	38.93
$k=4$	93.21	63.64	47.05
$k=8$	93.23	68.58	52.98
$k=12$	93.16	69.46	56.00
$k=16$	93.17	69.56	57.75
$k=32$	93.12	69.29	60.77

Table 4: Top-1, top-8 and top-32 overlap.

6 Conclusion

In this paper, we propose the Bi-directional Logits Difference (BiLD) loss, a novel optimization objective for distilling large language models (LLMs). The BiLD loss enhances distillation performance by filtering out long-tail noise and leveraging internal logits ranking information. It achieves superior distillation performance using only the top-8 logits compared to vanilla KL loss using full logits and other distillation methods. Our extensive experi-

ments across diverse datasets and model architectures confirm the effectiveness of the BiLD loss, demonstrating its ability to more efficiently capture key knowledge from the teacher model.

Limitations

Our approach falls within the realm of logits distillation, necessitating access to teacher logits. However, powerful LLMs such as GPT-4 (Achiam et al., 2023) and Gemini (Team et al., 2023) currently provide only output text or incomplete logits access, making our method unable to utilize these highly capable LLMs as teachers. Additionally, our Bi-directional Logits Difference (BiLD) loss requires shared vocabularies between the teacher and student models to ensure vector space alignment.

Another challenge lies in the computational complexity of our BiLD loss, particularly during the construction of logits differences using top- k logits. Although we demonstrate that using only the top-8 logits achieves better results than the vanilla KL loss, increasing the number of clipped logits leads to a rapid escalation in our method’s time overhead, which becomes a practical concern.

Furthermore, our approach directly clips the long-tail part of logits during distillation. While this approach improves performance, it unavoidably results in the loss of knowledge contained within the long-tail distribution. Investigating methods to better utilize the knowledge hidden in the long-tail distribution represents a promising avenue for future research.

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A Calculation Efficiency of BiLD

In Figure 4, we visualize the distillation speed of various methods during the distillation from Qwen-4B to 0.5B. Compared to the vanilla KL loss, our BiLD loss achieves better distillation performance with an acceptable increase in training time. Among all methods, DKD (Zhao et al., 2022) and NKD (Yang et al., 2023b), which are designed for vision models, have the slowest computation speeds due to the calculation of numerous intermediate variables. In contrast, the computation speeds of RKL, NormKD, and top- k KL are comparable to the vanilla KL loss.

In the code implementation, the BiLD loss consists of two main steps: selecting the top- k logit values and calculating the internal pairwise differences. Our analysis reveals that the latter step

is where the significant time expenditure occurs. The time complexity for computing the internal pairwise differences is $\mathcal{O}(n^2)$, and it frequently necessitates extracting values from the tensor. This has become the time bottleneck for the BiLD loss.

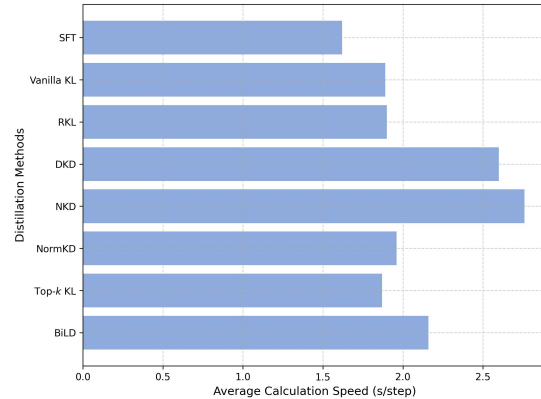


Figure 4: The average calculation speed of different distillation methods.

B Details about Dataset Sizes

Details about the dataset sizes are shown in Figure 5. There are significant size differences among the datasets, with the smallest datasets (CB, COPA, WSC) differing by three orders of magnitude from the largest dataset (ReCoRD).

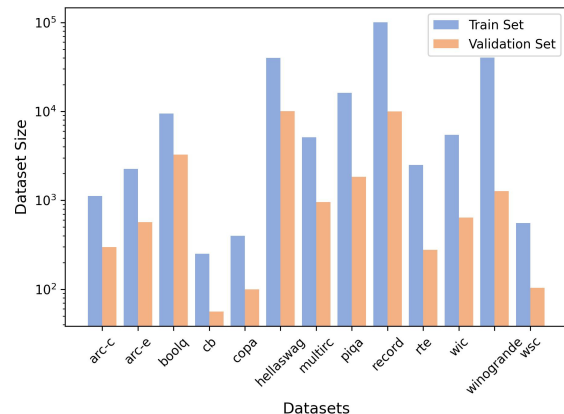


Figure 5: A visualization of the dataset sizes.

C Toy Experiment to Compare Vision Model and LLMs’ Logits

The five images we used in the toy experiments are shown in Figure 6, and the five sets of instructions are in Table 5.



Figure 6: Five images used in the toy experiment. All these images are under the CC0 license, meaning they can be used for scientific purposes without risk.

	Instructions Content
Instruction 1	Question: A mass of air is at an elevation of 1000 meters in the low pressure center of a Northern Hemisphere storm. Which of the following best describes the motion of air particles in this air mass due to storm conditions and the rotation of Earth as the air mass moves outward? Choices: ['Air particles move up and to the left.', 'Air particles move up and to the right.', 'Air particles move down and to the left.', 'Air particles move down and to the right.'] Answer:
Instruction 2	Premise: A: No, I don't either. B: Uh, I mean it's, you know it, A: I don't think it's going to change very much Hypothesis: it's going to change very much Question: Determine whether the premise entails the hypothesis or not. Choices: ['entailment', 'neutral', 'contradiction'] Answer:
Instruction 3	Goal: Keep laptop from overheating. Choose the most sensible solution to achieve the goal. Choices: ['Use on top of egg carton.', 'Use on top of egg shells.'] Answer:
Instruction 4	Choose the most sensible text to replace the "_" in the following sentence: Kyle asked Brett for some tips on healthy eating because _ has recently lost weight. Choices: ['Kyle', 'Brett'] Answer:
Instruction 5	Meanwhile, in the forest, the elephants are calling and hunting high and low for Arthur and Celeste , and their mothers are very worried. Fortunately, in flying over the town, an old marabou bird has seen them and come back quickly to tell the news. Question: In the above text, does 'their' refer to 'their mothers'? Choices:['true', 'false'] Answer:

Table 5: Five instructions used in the toy experiment.

D Templates

The template of each dataset can be seen in Table 6.

Dataset	Template
Arc-C	Question: A scientist is measuring the amount of movement along a fault. Which tool is best used for making this measurement? Choices: ['barometer', 'stopwatch', 'meter stick', 'magnifying lens'] Answer:
Arc-E	Question: Which color shirt will reflect the most light on a hot, sunny day? Choices: ['black', 'blue', 'red', 'white'] Answer:
BoolQ	Turn on red – Right turns on red are permitted in many regions of North America. While Western states have allowed it for more than 50 years; eastern states amended their traffic laws to allow it in the 1970s as a fuel-saving measure in response to motor fuel shortages in 1973. The Energy Policy and Conservation Act of 1975 required in §362(c)(5) that in order for a state to receive federal assistance in developing mandated conservation programs, they must permit right turns on red lights. All 50 states, the District of Columbia, Guam, and Puerto Rico have allowed right turns on red since 1980, except where prohibited by a sign or where right turns are controlled by dedicated traffic lights. (On January 1, 1980, Massachusetts became the last US state to allow right turns on red.) The few exceptions include New York City, where right turns on red are prohibited, unless a sign indicates otherwise. Question: is it legal to turn right on red in california? Choices: ['true', 'false'] Answer:
CB	Premise: B: And I've worked in the hospital for fifteen years and I've taken care of a few AIDS patients. A: Uh-huh. B: Uh, when they asked us did we want to, uh, keep it the same or, uh, spend more, spend less, uh, I think right now what they're spending is adequate. Uh, for my personal opinion. Uh, because I think it's something that's going to take them a while to come up with a, uh, vaccine for. A: Yeah. Uh-huh. Uh-huh. B: I don't think it's going to be that easy to come up with Hypothesis: it is going to be that easy to come up with Question: Determine whether the premise entails the hypothesis or not. Choices: ['entailment', 'neutral', 'contradiction'] Answer:
COPA	Premise: The woman betrayed her friend. Question: What could be the possible effect of the premise? Choices: ['Her friend sent her a greeting card.', 'Her friend cut off contact with her.'] Your answer:

HellaSwag	Please choose the most appropriate text to complete the passage below: Passage: [header] How to clean a plastic retainer [title] Rinse the retainer with warm or cold water. [step] The water will prep your retainer for the cleaning process. [title] Apply a mild soap onto a toothbrush. Choices: ['[step] Rinse the retainer under the faucet bowl with warm water. Suds will accumulate on the toothbrush.', '[step] Rinse the retainer slowly from top to bottom and then wipe it on the toothbrush. Soap can effectively clean a plastic retainer but it can potentially cause irritation.', '[step] If you are using an old toothbrush, you may brush the bristles for pleasure. Fill a bucket, then fill it with a cup of liquid soap.', '[step] You can use liquid castile soap or a mild dishwashing detergent. Additionally, use a soft-bristled toothbrush.'] Answer:
MultiRC	Passage: One of the most dramatic changes in priorities proposed by the City Council would shift \$25.6 million from funding for court-appointed lawyers to the Legal Aid Society. In a document released yesterday to justify its reordered priorities, the Council contended that Legal Aid can achieve greater economies of scale than lawyers appointed pursuant to Article 18-B of the County Law. The Council document also noted that inexplicably 18-B lawyers are handling 50 percent of the indigent criminal cases in New York City, even though their mandate is to handle only multi-defendant cases where the Legal Aid Society had a conflict. In past years, the City Council had consistently added \$5.6 million to the \$54.7 million proposed for the Legal Aid Society by former Mayor Giuliani, bringing the total to just a shade over \$60 million. But this year for the first time, the Council is proposing shifting more than \$20 million in funds earmarked by the Mayor for 18-B lawyers to the Legal Aid Society, which would increase its total funding to \$80.4 million. That would reflect a jump in its current finding of about one-third. Meantime, the City Council proposed slashing the Mayor's allocation of \$62.8 million for 18-B lawyers by 66 percent, to \$21.4 million. Question: By increasing current funding to the Legal Aid society by \$25.6 million, how much is the Council increasing their funding? Choices: ['\$60 million', '\$62.8 million', 'One third', '\$54.7 million', '\$80.4 million'] Note: 1. there can be multiple correct answers. 2. each line contains one answer. 3. If no correct answer, reply 'none'. Your answer:
PIQA	Goal: how do you flood a room? Choose the most sensible solution to achieve the goal. Choices: ['fill it with objects.', 'fill it with water.'] Answer:

ReCoRD	A father has admitted killing his 13-year-old son by giving him a morphine tablet when the boy complained that he was feeling ill. Kevin Morton gave his son Kye Backhouse an extremely strong painkiller, a court heard - a mistake which he says he will 'have to try and live with it for the rest of my life'. He could now face jail after pleading guilty to manslaughter over the teenager's death at Preston Crown Court. Tragedy: Kevin Morton, right, has admitted killing his son Kye Backhouse, left, by giving him morphine 'Happy-go-lucky' Kye was found dead at his home in Barrow-in-Furness, Cumbria in October last year.@highlight Morton gave Kye Backhouse a strong painkiller when he was ill@highlight teenager subsequently died and his father has admitted manslaughter@highlight, 49, faces jail when he is sentenced next month Question: Death: @placeholder, 23, complained of feeling unwell before his father gave him the strong painkiller What is the @placeholder?" Answer:
RTE	Premise: Euro Disney is one of the most popular theme parks of USA. Hypothesis: Euro-Disney is an Entertainment Park. Question: Determine whether the premise entails the hypothesis or not. Choices: ['entailment', 'not_entailment'] Answer:
WiC	Sentence1: An early movie simply showed a long kiss by two actors of the contemporary stage. Sentence2: We went out of town together by stage about ten or twelve miles. Question: Does 'stage' have the same meaning in both sentences? Choices: ['true', 'false'] Answer:
WinoGrande	Choose the most sensible text to replace the '_' in the following sentence: Natalie was less religious than Patricia, therefore _ attended church services more often on Sundays. Choices: ['Natalie', 'Patricia'] Answer:
WSC	The mothers of Arthur and Celeste have come to the town to fetch them. They are very happy to have them back, but they scold them just the same because they ran away. Question: In the above text, does 'them' refer to 'mothers'? Choices:['true', 'false'] Answer:

Table 6: The template of each dataset.